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CLIMATE-SMART LAND AND WATER MANAGEMENT FOR SUSTAINABLE AGRICULTURE†

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ABSTRACT

Indian agriculture is vulnerable to climate variability and change. The Coupled Model Inter-comparison Project Phase 5 (CMIP5) based model ensemble projects a warming of 1.5–4.3°C and an increase in precipitation by 6–14% for India under different representative concentration pathways (RCPs) for the 2080s (2071–2100) compared to the baseline period (1961–1990). The Indian Council of Agricultural Research through its network project on National Innovations on Climate Resilient Agriculture (NICRA) has demonstrated different land and water management interventions in the vulnerable districts of India to enhance the resilience of agriculture to climate change and variability. Rainwater harvesting and its judicious utilization through farm ponds and community tanks in rainfed districts demonstrated the ability to build resilience and also enhance cropping intensity (20–135%). Construction of temporary check dams (*bori-bandhan*) in high-rainfall areas brought lowland areas into cultivation which otherwise would have remained fallow due to inundation in the *kharif* (monsoon) season. Renovation of *aahar* (water reservoirs) and construction of farm ponds in Bihar ensured protective irrigation during dry spells in *kharif*, increasing paddy productivity by 20.7%, and also raised the groundwater level by 30 cm. Adoption of participatory integrated watershed management within a river basin perspective provides a framework for climate change adaptation through reduced runoff, increased groundwater recharge, improved crop productivity and increased carbon sequestration. Copyright © 2017 John Wiley & Sons, Ltd.

KEY WORDS: climate change; NICRA; technology demonstration component; NMSA; *bori-bandhan*; *aahar*; *pynes*

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RÉSUMÉ

L'agriculture indienne est vulnérable à la variabilité climatique et au changement climatique. La phase 5 du projet d'intercomparaison de modèles couplés (CMIP5) prévoit un réchauffement de 1.5 à 4.3°C et augmente les précipitations de 6 à 14% pour l'Inde sous différentes voies représentatives de concentration (RCP) pour les années 2080 (2071 à 2100) par rapport à la période de référence (1961–1990). Le Conseil indien de la recherche agricole à travers son projet de réseau sur les innovations nationales sur l'agriculture résiliente au climat (NICRA) a validé différentes interventions de gestion des eaux et des terres dans des districts indiens vulnérables afin d'accroître la résilience de l'agriculture aux changements et variabilités climatiques. La récolte de l'eau de pluie et son utilisation judicieuse à travers les étangs à la ferme et des bassins communautaires dans les districts pluviaux ont démontré leur capacité à créer de la résilience et tout en renforcer l'intensité des cultures (20 à 135%). La construction de barrages à contrôle temporaire (*bori-bandhan*) dans les zones à forte pluviométrie a permis de cultiver des terres basses qui, autrement, seraient restées en jachère en raison de l'inondation dans la saison de *kharif* (mousson). La rénovation de réservoirs d'eau (*aahar*) et la construction d'étangs à la ferme dans Bihar ont permis une irrigation protectrice pendant les périodes de sécheresse au *kharif*, augmentant la productivité du paddy de 20.7% et augmentant également le niveau d'eau souterraine de 30 cm. L'adoption d'une approche participative intégrée de la gestion des bassins hydrographiques fournit un cadre pour l'adaptation au changement climatique grâce à une réduction du

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ruissellement, une augmentation de la recharge des eaux souterraines, une amélioration de la productivité des cultures et une augmentation de la séquestration du carbone. Copyright © 2017 John Wiley & Sons, Ltd.

MOTS CLÉS: changement climatique; NICRA; composant de démonstration technologique; NMSA; *bori-bandhan*; *aahar*; *pynes*

INTRODUCTION

Agriculture is the sector most sensitive to, and greatly influenced by, climate change and variability. The Intergovernmental Panel on Climate Change (IPCC) in its Fifth Assessment Report observed that climate change is already happening in South Asia and impacts are already being felt (IPCC, 2014). With rising temperatures and changes in precipitation patterns due to climate change, it is likely to have serious implications for water resource availability, thereby affecting agriculture. Different models project a warming of 1.5, 2.4, 2.8 and 4.3°C for India under the RCP2.6, RCP4.5, RCP6.0 and RCP8.5 scenarios, respectively, during the 2080s (2071–2100), compared to the baseline period of 1961–1990 (Chaturvedi *et al.,* 2012). Similarly, all-India precipitation is projected to increase by 6, 10, 9 and 14% during the 2080s under the RCP2.6, RCP4.5, RCP6.0 and RCP8.5 scenarios, respectively. In comparison to the pre-industrial base of the 1880s (1861–1900), the all-India annual mean ambient temperature is projected to increase by 1.7–2.0°C by the 2030s and by 2.0–4.8°C by the 2080s, while precipitation is projected to increase 1.2–2.4% by the 2030s and 3.5–11.3% by the 2080s. As Indian agriculture is monsoon dependent, any change in frequency and magnitude of rainfall will be reflected in water availability both in irrigated and rainfed areas.

The projected increase in droughts, cyclones, extreme precipitation events and heatwaves will result in greater instability in food production. A reduction in the number of rainy days, heavy rainfall events, late onset of the monsoon, mid-season droughts, etc. have been witnessed in recent years, adversely affecting the crop production. At country level, during the last 15 years, 7 years recorded lower than normal rainfall, 6 years recorded more than normal rainfall and 2 years recorded average rainfall (Srinivasa Rao *et al*., 2015). Analysis of rainfall data of 2001–2015 revealed that the monsoon rainfall deficit was more than 10% in 6 years, i.e. 2001, 2002, 2004, 2009, 2014 and 2015, with maximum deficit being 23% in 2009 followed by 19% during 2002. The increased incidence of extreme weather events (Table I) as witnessed in recent decades poses serious challenges to sustaining agriculture and this could be attributed to climate change.

In general, there has been a decline in production over previous years during drought years (Table II). Production data on food grains, pulses, cereals and nutritious cereals indicate that despite higher deficits of rainfall, food grain production has stabilized, which could be due to stabilization of assured canal regions and increased use of groundwater. Pulse production continued to be dependent on rainfall. Though there is a general trend of a decrease in pulse production commensurate with deficit rainfall, spatial distribution across the country plays an important role. A similar trend could also be observed with nutritious cereals. Total cereal production which includes paddy and wheat along with nutritious cereals is relatively stable despite a decrease in rainfall, indicating the preference for paddy and wheat cultivation in canal commands and using groundwater to a large extent. Floods are also a recurring phenomenon in India. Twenty-three of the 36 states and union territories in the country are subject to floods and 40 million ha of land, roughly one-eighth of the country's geographical area, is prone to floods (Sikka *et al.*, 2016a).

Table I. Monsoon rainfall and extreme weather events during 2001–2015 in India

Year	SWM rainfall departure from normal (%)	Extreme events
2001	−15	Drought
2002	−19	20-day heatwave during May in Andhra Pradesh
2003	+2	Extreme cold winter in 2002–03
2004	−13	Abnormal temperatures during March 2004; drought-like situation in India in July 2004
2005	−1	Flood in 2005
2006	−1	Cold wave during 2005–06; floods in arid Rajasthan and Andhra Pradesh, and drought in north-eastern regions in 2006
2007	+5	Abnormal temperatures during third week of January to first week of February 2007
2008	−2	–
2009	−23	Severe drought all over India
2010	+2	One of the warmest years
2011	+1	Failure of September rains in Andhra Pradesh
2012	−8	Drought in Punjab, Haryana, Gujarat and Karnataka. Cyclone, floods in Andhra Pradesh
2013	+6	Flood in Uttarakhand, cyclone *Phalin* and hailstorms
2014	−12	Flood in Jammu and Kashmir, cyclone *Hudhud*
2015	−14	Extreme heatwaves, drought

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Table II. Food grain production with rainfall variability—all India scenario (deviation from normal scenario)

Year	Departure from normal (%)				
	Rainfall	Food grains	Pulses	Total cereals	Nutritious cereals
2002–2003	−19	−13.3	−15.6	−13.1	−20.8
2009–2010	−23	−4.8	−4.1	−4.8	−12.6
2012–2013	−8	0.6	1.8	0.5	−5.4
2014–2015	−12	−1.4	−0.8	−1.1	1.3
2015–2016	−14	−1.3	−8.6	−0.8	−10.4

CLIMATE CHANGE, WATER AND AGRICULTURE

India is more vulnerable to climate change in view of the dependence of its huge population on agriculture, excessive pressure on natural resources and poor coping mechanisms. Around 121 million ha (Mha) of the total geographical area (36.5% of the total) of the country are affected by various kinds of land degradation (National Academy of Agricultural Sciences (NAAS), 2010). Per capita availability of arable land has declined from 0.34 ha in 1961 to 0.13 ha in 2012 (Food and Agriculture Organization of the United Nation's statistical database (FAOSTAT), 2016). Per capita annual availability of water has declined from 5180 m^3 in 1951 to 1540 m^3 in 2011 (Central Water Commission (CWC), 2015) and is expected to reduce to 1340, 1150 and 1140 m^3 by 2025, 2050 and 2065, respectively (Jain, 2011). The anticipated impact of climatic change and climate variability including seasonality problems is likely to lead to increased pressure on already scarce water resources. Simulation studies on the impact of climate change on hydrology and water resources have shown a varied response in Indian river basins depending upon the hydrological model, climate change scenarios and downscaling approaches used (Gosain *et al*., 2011; Islam *et al*., 2014). An increase in precipitation intensity under changing climate may accelerate the soil erosion process; increased sea level may result in seawater intrusion, aggravating land degradation processes.

In spite of having more than 65 Mha covered under irrigation, Indian agriculture is monsoon dependent. The summer monsoon season spanning over 4 months (June–September) contributes about 75% of the annual precipitation. With increasing interannual variability in summer monsoon rainfall, intra-seasonal variability, large-scale spatial variability and occurrence of frequent extreme weather in the future under changing climate scenarios, climate change may affect the agricultural production system by intensifying abiotic and biotic stresses which influence the germination, growth, reproduction, pollination, fertilization and maturity processes of crops, as well as crop duration and incidence of diseases and pests, enhanced photosynthesis and water use efficiency, limiting water availability for irrigation, and enhancing frequency of extreme weather events (Sikka *et al*., 2016b). Scarcity of irrigation water under changing climate scenarios is a great concern for rice–wheat production systems, which cover 13.5 Mha in the Indo-Gangetic Plain (IGP) and are vital for food security (Ladha *et al*., 2003). The declining trend in grain yields of rice and wheat in the IGP have been attributed partly to weather changes (Aggarwal *et al*., 2004). Simulation studies have shown a decrease in yields of irrigated and rainfed rice (*Oryza sativa* L.) (Naresh Kumar *et al*., 2013), wheat (*Triticum aestivum* L.) (Naresh Kumar *et al*., 2014) and *kharif* (monsoon) maize (*Zea mays* L.) (Byjesh *et al*., 2010) under future climate change scenarios. While adoption of simple agronomic adaptation strategies may offset negative impacts in the near future, development of alternative adaptation technologies are essential in the longer run.

As water is the medium of expression of the impacts of climate change, land and water management will be at the centre of climate change adaptation strategies. With shrinking land and water resources, improving land and water productivity is critical for ensuring food, nutritional and livelihood security. Challenges of water scarcity could effectively be addressed through adoption of smart water management interventions and modern irrigation technologies to enhance water productivity by involving a combination of area-specific approaches for both supply- and demand-side management. In this paper, experiences of some of climate-smart land and water management interventions are briefly discussed.

PORTFOLIO OF CLIMATE-SMART LAND AND WATER MANAGEMENT FOR RESILIENT AGRICULTURAL SYSTEMS

Climate-smart agriculture (CSA) deals with multiple aspects of climate change for transforming and reorienting agricultural systems to support food security under the new realities of climate change (FAO, 2010; Campbell *et al*., 2014; Lipper

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et al., 2014). CSA aims to achieve three objectives: (i) sustainably increasing agricultural productivity to support increased incomes, food security and development; (ii) adapting and building resilience to climate change from the farm to national levels; and (iii) decreasing greenhouse gas emissions and increasing carbon sinks from agriculture compared with past trends. Although it requires consideration of all three objectives to derive locally acceptable solutions, the relative importance of each objective varies across locations and situations. CSA pathways which promote (i) coordinated policymaking; (ii) strengthened institutions; (iii) evidence-based planning; (iv) stable dedicated financing result in higher resilience and lower risks to food security, whereas business as usual leads to higher risks of food security and lower resilience of food and agricultural systems (Lipper *et al*., 2014). There are several land and water management practices such as watershed management, water harvesting and its efficient utilization using precision and smart irrigation practices, conservation agriculture, etc. that sustainably increase productivity, enhance resilience to climatic stresses, reduce greenhouse gas emissions and thus appear to be promising CSA options. Expanding the area under irrigated agriculture or adoption of more efficient water management practices can enhance carbon storage in soils through enhanced yields and residue returns (Lal, 2004).

Climate-resilient agriculture refers to agricultural practices that enhance the capacity of the agro-ecosystem to relieve, resist, respond and recover from climate-related disturbances. This essentially involves judicious management of natural resources, viz. land, water, soil and genetic resources, through adoption of best bet practices. Different natural resource management interventions such as *in situ* moisture conservation, biomass mulching, residue incorporation, brown and green manuring, water harvesting and recycling for supplemental irrigation, improved drainage in flood-prone areas, conservation tillage wherever appropriate, artificial groundwater recharge and water-saving irrigation methods have the potential to increase resource-use efficiency and increase resilience in agricultural production systems. Similarly, different agronomic management practices such as adjustment of planting dates and the cropping calendar, adoption of stress-tolerant crop varieties, crop rotations, intercropping, integrated weed and pest management, integrated nutrient management (INM), site-specific nutrient management (SSNM), agroforestry, etc. are other important components suggested for strategic adaptation to climate change and variability in India. Some of the adaptation measures such as adoption of micro-irrigation technology, intermittent irrigation (alternate wetting and drying) of paddy fields, resource-conserving technologies (RCTs) such as zero or minimum tillage (with or without crop residues), INM and SSNM also have the co-benefit potential to mitigate the effects of climate change.

The identification, prioritization and promotion of available CSA technologies considering local climatic risks are major challenges in planning and implementation of CSA practices. Hochman *et al.* (2017) suggested participatory action research with smallholder farmers, coupled with field testing and simulation analysis to produce practical, sustainable and productive climate-smart adaptation options. Table III presents some important interventions and indicators thereof that can be visualized as smart land and water management practices together with production practices for resilient agriculture systems.

The Indian Council of Agricultural Research (ICAR) initiated the National Initiative on Climate Resilient Agriculture (NICRA) in 2011 (renamed as National Innovations in Climate Resilient Agriculture in 2014) to generate climate-resilient agricultural technologies. The project aims at enhancing the resilience of Indian agriculture to climate change and its variability and it operates through four major components, namely strategic research, technology demonstration, capacity building and sponsored/competitive grants for basic research. Under the technology demonstration component (TDC) of NICRA, area-specific available smart practices have been demonstrated through a farmers' participatory mode with a major focus on natural resource management in 121 climatically vulnerable districts across the country (Figure 1). It is implemented through Krishi Vigyan Kendras (KVK) farm science centres with the aim of building resilience of the farming community to droughts, floods, cyclones, unseasonal rains, heat and cold waves. Farmers in NICRA villages adopted a multi-pronged approach such as creation/renovation and/or desiltation of community tanks, check dams, desiltation of conveyance systems for safe disposal of runoff into community tanks, farm ponds and *in situ* conservation of rainfall through land and water management practices. The experiences gained by NICRA villages have opened up opportunities for upscaling the successful experiences in other parts of the districts through various developmental programmes. Evidence of enhanced resilience to cope with adverse climate conditions based on NICRA experiences is discussed in the following section.

EVIDENCE OF IMPROVED RESILIENCE ARISING FROM SMART LAND AND WATER MANAGEMENT INTERVENTIONS

Soil moisture security and enhanced water storage through water harvesting in surface and subsurface storage structures at field/farm/village/watershed level have proved vital in improving resilience at the local level for farming systems. This has been especially much more important in low to medium rainfall zones where rainfall variability and occurrence of intense rain events are high.

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Table III. Climate-smart agricultural practices for land and water management

Module	Intervention	End user	Outcome
Moisture deficit and/or drought management	*In situ* soil moisture conservation practices, including land modification, raised bed and furrows, conservation furrows, broad bed furrow, mulching	Individual	On-farm water conservation and improved productivity
	Farm ponds	Individual	Improved productivity, enhanced water availability, diversified enterprises, increased cropping intensity
	Tanks	Community	
	Check dams	Community	
	Resource conservation technology (RCTs), conservation agriculture	Individual/ community	Improved productivity, improved livelihood
	Recharging wells, percolation ponds	Individual	Enhanced water availability, productivity
	Contingency crop planning	Individual/ community	Increased resilience, improved productivity
	Planting methods	Individual	Increased resilience, improved productivity
	Biomass/residue recycling	Individual	Soil productivity maintenance, increased resilience, improved productivity
	Adjustment in planting time of *rabi* (winter) crops	Individual	Water saving, increased productivity
Floods, cyclones, excess and unseasonal rains	Land shaping in coastal tracts prone to cyclones, excess rainfall. Sea water intrusion	Individual	Sustainable increase in cropping intensity
	Improving drainage/conveyance efficiency in flood-prone areas	Community	Increased productivity and resilience
	Flood-tolerant varieties	Individual	Increased resilience
	Integrated farming system models	Individual	Increased resilience through year- round farm income
	Planting methods (broad bed furrow, ridge-furrow, raised bed and sunken furrow system, etc.)	Individual	Improved productivity and resilience

Community tanks/ponds for augmenting village-level water resources

Community tanks, traditionally recognized as resilient systems, silt up over the years and have become defunct due to poor maintenance and neglect. Desiltation of these tanks helped increase/regain surface water storage, as well as increased groundwater recharge. The spin-off effects of farmers spreading the rich silt deposited in these structures in their fields helped to improve soil water-holding capacity. An increase in the yield of cotton and castor (15–18%) was observed in silt-applied fields in Kurnool districts of Andhra Pradesh over the check due to improved fertility status. Desiltation and renovation of three community ponds with a surface area of 1050, 5040 and 3610 m^2 and storage capacity of 5740, 23 000 and 16 500 m^3 in Tamil Nadu resulted in an additional increase of water storage of 36 600 m^3, as well as recharging of groundwater. This additional water helped in the cultivation of crops over an area of 138 ha with diversified crops in the villages of Jambumadai and Vadavathur in Namakkal district (Table IV).

Renovation of old *aahar* (water reservoir) and *pyne* (conveyance channel) systems and construction of farm ponds in Nawada district of Bihar helped to harvest an additional 20 000 m^3 of water after renovation. This helped to provide protective irrigation for 24 ha during dry spells in *kharif* which resulted in enhancing the productivity of paddy by 20.7%, increasing the groundwater level by 30 cm and making drinking water available for livestock.

In another example, renovation and deepening of irrigation channels (Mentepudi channel and VWS channel) in West Godavari district not only helped in providing irrigation to farmers at the tail end but also helped in efficient and timely disposal of rainwater by avoiding flooding and submergence of crops at times of intense rainfall events. During the *Neelam* cyclone, flood water up to a height of only 42 cm in the command area was observed, whereas in other untreated areas flood water was up to 122 cm and the crop was completely submerged. This resulted in avoiding a yield loss and realized paddy productivity of a level of 2630–4130 kg ha^{-1} under the renovated channels, whereas yield was only 1500 kg ha^{-1} in untreated areas that suffered submergence.

Farm ponds (individual/small group based)

Supplemental irrigation during dry spells from farm ponds has long been identified as an adaptation strategy to rainfall

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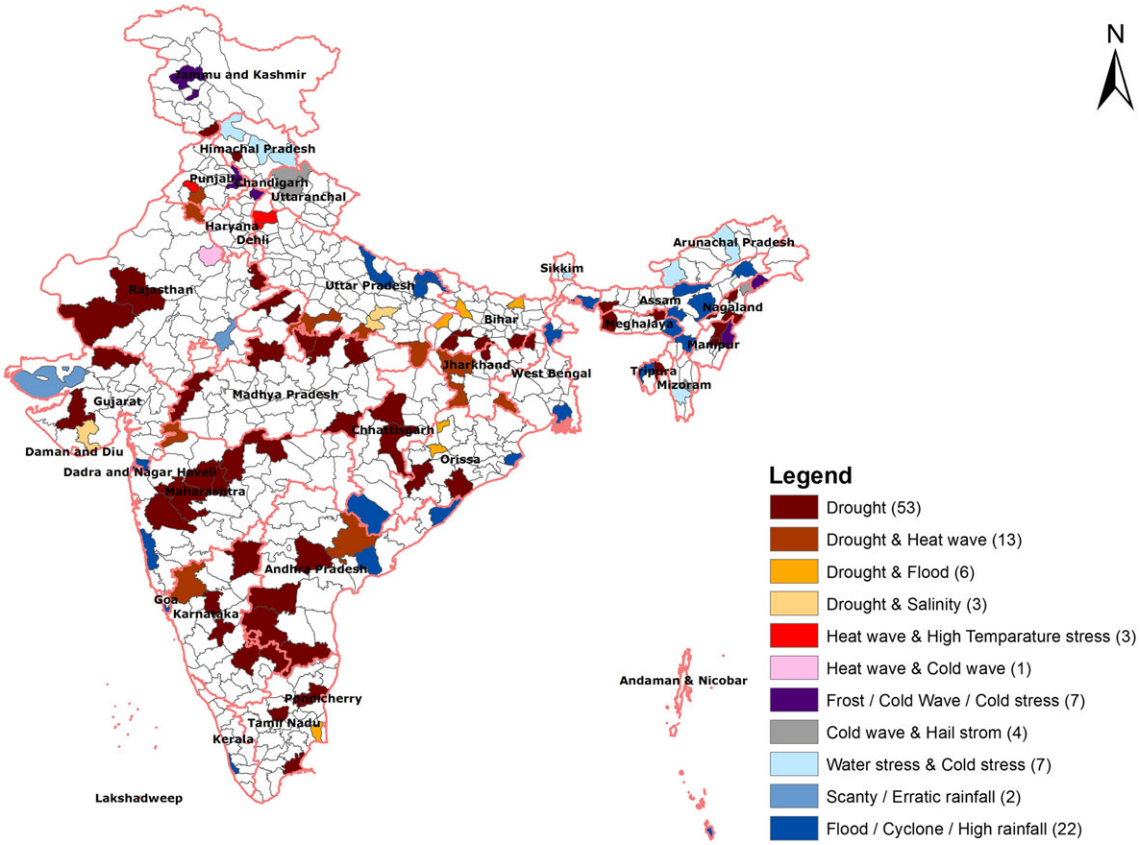


Figure 1. Climatologically vulnerable districts of India. [Colour figure can be viewed at wileyonlinelibrary.com]

Table IV. Effect of desiltation of community ponds

Name of the pond	Dimensions (m × m × m)		Groundwater recharge	Area irrigated (ha)	Major cultivated crops	
	Before	After			Before	After
Senguttai	36 × 26 × 3.65	38 × 28 × 5.4	18 open wells and 11 bore wells	14	Onion and sorghum	Onion, groundnut and sorghum
Aayiramkuttai	30 × 22 × 1.85	92 × 55 × 4.5	45 open wells and 112 bore wells	74	Onion and sorghum	Onion, paddy and groundnut
Periyakalingi-kuttai	49 × 22 × 3.65	112 × 32 × 4.5	45 open wells and 156 bore well	138	Onion and sorghum	Onion, groundnut and sorghum

variability, including climatic change to overcome water stress and ensure food security. Various models of small-scale water-harvesting systems have been promoted by governmental and non-governmental organizations involving different pond sizes, lining material and promotion of reuse of harvested water for different crops at critical crop growth stages. Farm ponds have been widely adopted as one of the key interventions in NICRA villages across different rainfall zones. The majority of farmers opted to cultivate vegetables with harvested water at a ratio of 1: 10 (command to catchment area) with sustained profits. This led to an increase in irrigated area, crop productivity and increased groundwater level. More importantly, farmers having farm ponds were able to cope with moisture stress during dry spells using supplemental irrigation, and in some cases even cultivated two crops a year during high-rainfall years and received a higher income (Table V). Further, adoption of pressurized irrigation systems (drip and sprinkler) resulted in efficient utilization of harvested rainwater and the benefit–cost ratio ranged from 2.0 to 3.9 (Table VI).

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Table V. Examples of successful interventions with efficient use of harvested rainwater for crop production during deficit rainfall

Village	KVK/ district	No. of farm ponds	Storage capacity (m^3)	No. of farmers covered	Irrigation potential created (ha)	Increase in cropping intensity (%)
Kukurha	Buxer	1	1 800	6	1.75	100
Affaur	Saran	2	11 500	42	10	25
Chopanadih	Koderma	1	16 560	50	20	40
Takli	Amravati	2	8 400	25	10	20
Nacharam	Khammam	4	1 460	4	9.6	40
Sanora	Datia	3	9 500	10	19	100
Nagenhalli	Tumkur	2	2 500	25	65	135
Nagenhalli	Tumkur	13	22 100	25	12	100

Table VI. Effect of pressurized irrigation as adaptation measure

Village	KVK/ district	Technology demonstrated	Crop	Area (ha)	Net return (Rs)	Benefit– cost ratio
Lagga	Chamba	Sprinkler	Cauliflower	2	380 000	3.9
Kukurha	Buxer	Sprinkler	Rice	4	27 300	2.0
Affaur	Saran	Micro-irrigation	Long gourd	2	66 000	3.0
Takli	Amravati	Sprinkler	Cotton	10	36 300	2.5
Nacharam	Khammam	Drip	Chilli	2	9 000	3.0
Nidhan	Morena	Sprinkler	Wheat	8	52 200	2.6
Vadavathur	Namakkal	Sprinkler	Onion	35	216 000	3.2
Nagenahalli	Tumkur	Drip	Brinjal	2	103 000	3.7
Nagenahalli	Tumkur	Sprinkler	Groundnut	1	59 500	3.3

Jalkund, low-cost miniature rainwater-harvesting ponds (size 5 × 4 × 1.5 m) with a silpaulin lining having a storage capacity of 30 000 l for harvesting rainwater during the rainy season and its subsequent use during dry periods to provide life-saving irrigation to high-value winter vegetables, have been successfully adopted in the hilly north-eastern region of the country as a smart water management intervention. In East Sikkim, a *Jalkund* of size 5 × 4 × 1.5 m with a plastic lining, harvested about 30 000 l of rainwater, which was judiciously utilized by four farmers over an 1.8 ha area as life-saving irrigation through micro-irrigation for sole cropping, vegetable seedlings and potato. Judicious use of *Jalkund* water to irrigate *rabi* crops during dry spells with sprinklers made vegetable production a profitable venture for many farmers.

Increased in-stream storage: Check dams in rainfed areas

Ex situ storage of water in seasonal streams at suitable sites is an important strategy to store runoff water in different rainfall zones to provide resilience for direct use of surface storage or for improving groundwater availability. In the majority of NICRA villages, check dams (new/desilting of existing ones), low-cost temporary check dams (sand bag check dams), and *Bori-bandhans* (polybag check dams) were major interventions in areas vulnerable to drought. These structures not only increased water storage capacity but also increased groundwater recharge. For instance, with the construction of five *Bori-bandhans* in the Datia district of Madhya Pradesh, 42 500 m^3 water were harvested. The harvested water could provide supplemental irrigation for 85 ha. Recharging of 31 open wells increased water availability for irrigation till March which was remarkable. Recharged open wells generated 20 ha of extra irrigation facility to the farmers in the village. Famers cultivated a mustard crop over 199 ha during the *rabi* season, resulting in a yield of 1280 kg ha^{-1} and boosting their net income to Rs. 29 200 ha^{-1} with a benefit–cost ratio of 3.1. Construction and desiltation of check dams in Pune district of Maharashtra increased the total water storage capacity from 47 800 to 165 000 m^3. This benefited the recharging of 96 wells located in the vicinity of these structures. The total area brought under cropping in the *kharif*, *rabi* and summer seasons as a result of this storage was about 49, 240 and 27 ha, respectively.

In situ soil moisture conservation

In situ rainwater management through the ridge and furrow method and a broad bed furrow system conserves rainwater at farm level and also drains excess water, which can be

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utilized for recharging groundwater. The conservation furrows, ridge and furrow, and broad bed furrow (BBF) systems were found to be effective in coping with moisture stress in the fields. This practice also helped to drain excess water from the fields to save the crop from waterlogging. This intervention is being widely promoted for cultivation of different crops in Maharashtra, Madhya Pradesh, Uttar Pradesh, Rajasthan, Odisha, West Bengal, Karnataka and Jharkhand. States such as Maharashtra have taken up BBF in a big way and support BBF through a subsidy scheme.

Soil and crop management practices

Different resource-conserving technologies (RCTs) such as zero or minimum tillage (with or without crop residues), bed planting of crops and direct-seeded rice have demonstrated substantial scope for improving irrigation efficiency and saving energy for groundwater pumping as well as reducing GHG emissions (Venkateswarlu *et al.,* 2012). Direct-seeded rice (DSR) and zero-till wheat (ZTW) generally gave a similar yield with less irrigation water and lower global warming potential (Figure 2).

Intermittent irrigation with a ponding depth of 5 ± 2 cm of water in rice fields 3 days after disappearance (DAD) of previously ponded water not only saves water but also reduces global warming potential by 25–30% over continuous flooding (Pathak, 2015; Chaudhari *et al.,* 2015). Micro-irrigation options are not only important for saving water but also relevant for saving energy (Shah, 2009) and reducing carbon emissions. Further, adoption of solar-powered drip systems results in water saving as well as energy saving and hence reduce carbon emissions. In eastern India, where groundwater is available at a shallow depth, solar pumps offer an ideal option for utilization of available groundwater with reduced emissions.

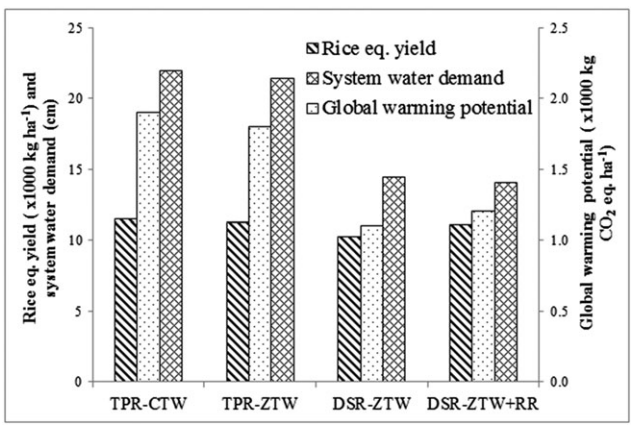


Figure 2. Effect of different resource conservation practices on irrigation water demand and global warming potential.

Watershed management—A framework for climate change adaptation

In rainfed areas, participatory integrated watershed management approaches covering biophysical and socio-economic interventions have paid rich dividends for ensuring resilience to climate variability and change. Management of natural resources at a watershed scale through a classical ridge to valley approach based on resource assessment interventions produces multiple benefits such as resource conservation, increased food production, improved livelihoods, protecting the environment, addressing gender and equity issues along with biodiversity concerns. Additionally, agroforestry interventions on marginal soils for soil conservation and restoration of degraded lands have the potential to increase soil carbon build-up and help mitigate and/or adapt to climate change. For instance, the examples of the Indo-German Watershed Development Programme (IGWDP) in Maharashtra by the Watershed Organization Trust (WOTR), Kadwanchi watershed in the Jalana district of Maharashtra, Hiware Bazar participatory watershed development model in the Ahmednagar district of Maharashtra, and other similar projects have demonstrated the importance of watershed management programmes in enhancing resilience and adaptation to climate change. These works as well as the impact studies of six watershed projects by Sharda *et al*. (2005) have demonstrated that resource conservation and management interventions have significantly augmented groundwater recharge, increased irrigated area and increased biomass production, resulting in flood moderation and drought management, employment generation and improvement in the socio-economic conditions of local people.

Several changes have been brought in the implementation of watershed programmes since their large-scale implementation from 1994 onwards. Some of these improvements include addressing the issues of synergy between neighbouring watersheds by enhancing the area to be considered for a watershed from 550 to 3000–4000 ha, a focus on water management and water budgeting, land use diversification, livelihood improvements, participation of local communities, implementation of programmes through local-level people institutions, etc. The sustainable rural livelihood programmes implemented by various research institutes under the National Agricultural Innovation Project helped to demonstrate and integrate the benefits of resource conservation activities to enhance resilience, realize better agricultural production and improve rural livelihoods (Srinivasa Rao *et al*., 2013; Rao *et al*., 2012). The watershed management programmes have acted as a source of convergence of many rural developmental activities in rainfed areas and have been recognized by the government of India as an important intervention to improve rainfed areas. The key element identified for success of watershed

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programmes is the participation of communities in local resource conservation activities in planning, designing, executing and maintaining. A watershed approach within a river basin perspective, integrating both top-down and bottom-up management approaches, provides the combined benefits of all the watersheds upstream and downstream of the river basin, enabling sustainable management of water resources. This very holistic concept integrating biophysical and social approaches could serve as a framework for climate change adaptation.

Co-managing floods and groundwater depletion at the river basin scale

As the frequency and intensity of floods are projected to intensify in the near future, a new management approach, Underground Taming of Floods for Irrigation (UTFI), has been demonstrated by the International Water Management Institute (IWMI) to tackle the dual challenges of floods and groundwater depletion in the Rampur district of Uttar Pradesh. UTFI strategically recharges aquifers upstream during periods of high flow, thereby preventing local flooding and simultaneously building resilience by providing additional groundwater for irrigation during the dry season for livelihood improvement. It involves diverting excess floodwater from canals and recharging aquifers through well-designed recharge structures placed in ponds and other depression areas (Pavelic *et al*., 2015), and evacuated through large-scale pump irrigation during the dry season.

CONCLUSIONS

The impact of climate change on agriculture will be one of the major factors influencing future food security. Adaptive responses are mainly related to technological interventions, management practices, sound governmental policy and political will to overcome the ill effects of climate change. The success of land and water management interventions implemented through the participatory process under NICRA across vulnerable districts in India has demonstrated the ability of such low-cost interventions to enhance resilience to climate change for sustainable agriculture. These have great potential for upscaling through supportive policies and programmes. Hence, apart from the use of technological advances to combat climate change, there has to be a sound and supportive policy framework. Governmental initiatives should be undertaken to identify and prioritize adaptation options in key sectors (early warning systems, water storage and diversion, etc). Some suggested policy interventions are:

- mainstreaming these area-specific climate-smart land and water management interventions through national and state development programmes: importantly, the Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), the National Mission for Sustainable Agriculture (NMSA), the National Water Mission and Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) and evaluating the impact of enhancing resilience to climate change and reducing global warming potential at a developmental scale. This will also accelerate achieving India's Intended Nationally Determined Contribution (INDC) target for 2030 by reducing GHG emissions, enhancing resource use efficiency and creating additional carbon sinks in farming systems, besides ensuring national food security;
- farm mechanization through custom hiring centres in promoting land management and promotion of micro-irrigation and solar pumps;
- exploring potential sites for diverting high water flows from rivers or canals when these flows pose a flood risk and recharging the groundwater via village ponds or small dams/check dams and recharge wells to reduce flood risks locally and improve groundwater conditions;
- drought preparedness and real-time implementation of contingency measures at field level through well-structured institutional support with strong government policy and convergence among various institutions;
- improved weather forecasting, hydrological monitoring, early warning systems, improved information and communication technology (ICT) and weather-based agro-advisory services for prevention and preparedness to cope with climate-related risks;
- village-level agricultural planning, based on land and water resources, to augment water availability for enhancing resilience through supply and demand management.

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